Solar Powered Terrain Walker
Team 1

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Introduction

The objective of the solar powered terrain walker is to make an autonomous solar powered robot that is capable of navigating difficult terrain and stands at a maximum of 18 inches tall. It also needs to carry a 8 oz payload. Our robot is 16 inches tall and has 4 legs.

Concept and Design

We constructed a walking robot that has 4 leg units, with the back two directly bolted to the chassis and the front two mounted via pivots and attached to the turning mechanism. Each leg unit is composed of the leg mechanism, motor, and encoder, all of which are mounted to the leg base and H bar. One additional motor is used to actuate the turning mechanism, giving the robot 5 motors. This, combined with the H bars minimizes weight and maximizes energy efficiency.

Legs

The final leg design is composed of a Chebyshev 4 bar mechanism driving a pantograph, which inverts and enlarges the 4 bar mechanism motion path. This design allows us to use one motor to actuate one leg, maximizing energy efficiency and in turn maximizing battery life. The motion path is flat for 270 degrees of the motor's rotation, with a quick return of 90 degrees of the crank's rotation. This motion path provides stable locomotion of the robot. This was originally designed in 2D, and the implementation in 3D was challenging, requiring us to develop the link configuration shown in Figure #1. Deep groove ball bearings were press fitted into the outside links to minimize friction.

Electronics

The robot consists of 5 continuous rotation 30 kg-cm servo motors that enable it to move. Servo motors provide the torque needed to lift the legs while rotating at a relatively slow rpm. The motor positions are tracked with an external encoder. The code uploaded to the Arduino enables the robot to move autonomously. A 7.4 volt 5200mAh lithium polymer battery supplies power to the motors and other electronics. A 20 watt solar panel with a built in controller is used to safely charge the battery.

Chassis and Steering

The chassis is comprised of PETG H beams that efficiently bear the weight of the robot while minimizing weight and space. The steering uses a 4 bar mechanism with one fixed linkage, as seen in Figure #6. A motor moves the linkage opposite of the fixed linkage. This allows rotation of the 2 side linkages with 1 motor. The leg modules are attached to the side linkages. The decision to use a 4 bar mechanism simplifies the design, reducing possible error and cost.

Manufacturing and Assembly

Each leg is composed of 7 links of which 3 were milled from aluminum and 4 were printed of PETG plastic. The joints connecting these links use shoulder screws and deep groove ball bearings with 4 and 9mm diameters respectively. This required difficult tolerancing, therefore green Loctite 609 was used for retention. Due to the elastic properties of PETG, bearings and fasteners were easily secured into the links. A 3D printed PETG H beam, along with brackets, bolts, and nuts were used to secure the leg units to the chassis. Gears were 3D printed for encoders to precisely track the motor's position.

Figures and Photos

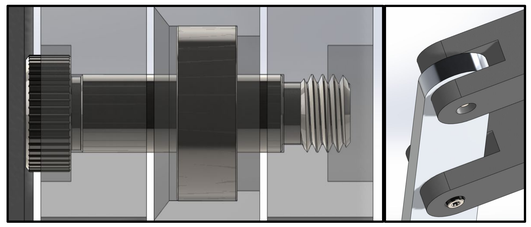


Figure #1: Link configuration

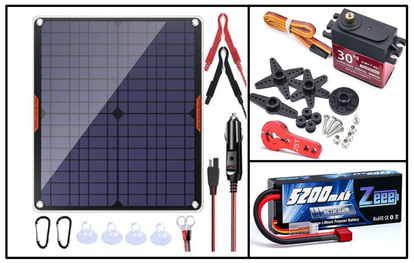
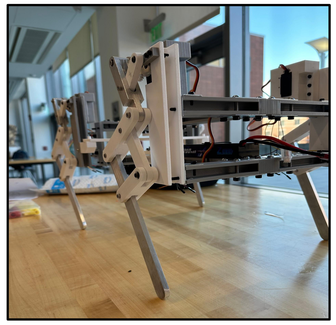


Figure #4: Solar panel, servo motor, and battery used in the robot



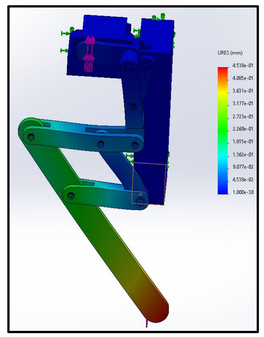


Figure #2: FEA deformation analysis on leg

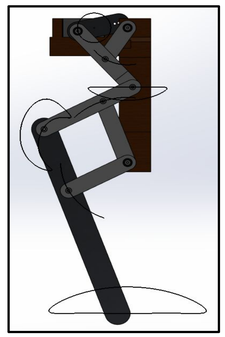


Figure #3: Leg motion path

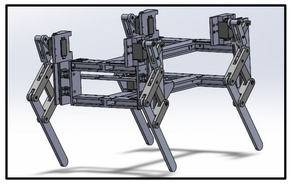


Figure #5: Solidworks model of robot

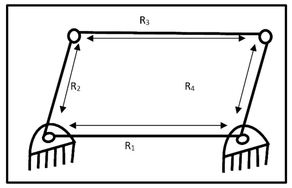
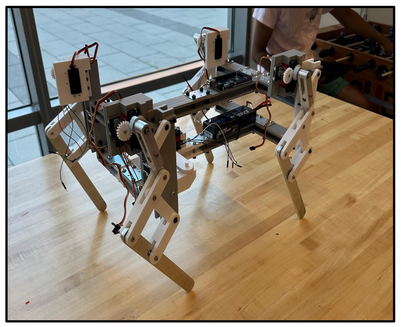


Figure #6: 4 Bar Steering Mechanism



Conclusion and Future Work

We have created a quadruped robot powered by a solar panel capable of navigating through terrain while carrying a small payload. The robot is designed to maximize energy efficiency. To achieve this, we used PETG for the chassis and the low stress portions of the legs. We also minimized the number of motors used, with one motor per leg and another for the turning mechanism. Deep groove ball bearings set on shoulder screws were used to minimize friction. Four of the seven links composing of the leg mechanism were PETG, while the high load links were made of 6061 Aluminum. The 20 watt solar panel combined with the high energy density lithium polymer battery allows the robot to recharge quickly and have a longer run time.

There are various ways the design can be optimized. For example, implementing a proper waterproof housing to protect the electronics and motors from water and debris would enhance its durability. Additionally, selecting motors with built-in encoders would simplify the manufacturing process.